

Farzaneh Fayazbakhsh

Bioinformatics Researcher | Single-Cell, Spatial Genomics & Molecular Biology

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Personal Statement

Interdisciplinary bioinformatics and molecular-biology researcher with hands-on experience in single-cell and spatial transcriptomics, sequencing workflows, cell culture and translational research. I use Python, R, Linux and HPC-based reproducible workflows to analyze high-dimensional biological data, and am motivated to connect gene-regulatory insights with experimental genomics and CNS repair research.

Bioinformatics & Computational Biology Projects

Automated Raw Sequencing Data Quality Control Pipeline

- Built an automated, parallelized Bash orchestration workflow designed for execution on High-Performance Computing (HPC) clusters via SLURM scheduling.
- Deployed FastQC and fastp dynamically inside isolated, version-controlled Singularity and Apptainer containers to automate raw sequence pairing, quality checking, and adapter trimming.
- Implemented multi-core processor allocation inside container runtimes to optimize throughput when executing string substitutions and batch file calculations.
- Orchestrated MultiQC container tasks to aggregate and output comprehensive data quality matrices, tracking processing adjustments across all sample cohorts simultaneously.

Tumor Microenvironment Profiling: scRNA-Seq & Spatial Transcriptomics

- Engineered end-to-end transcriptomic pipelines in R using Seurat and SCTransform to map high-dimensional gene expression across breast cancer tissue cohorts.
- Executed data pre-processing, Quality Control (QC), linear (PCA), and non-linear (t-SNE / UMAP) dimensionality reduction alongside graph-based clustering to resolve distinct cell populations and structural tissue features.
- Containerized both analysis environments using Docker and Docker Compose, ensuring full cross-platform compatibility and seamless runtime execution across varied infrastructure platforms.
- Managed pipeline documentation, software repositories, and strict workflow reproducibility utilizing Git and GitHub version control standards.

Computational Biology & Bioinformatics Skills

Programming & Pipeline Automation: Python, R, UNIX/Linux Shell, Nextflow (pipeline development), Git, and GitHub for version control

Genomics, High-Throughput Sequencing & Genetics: Practical experience in NGS data analysis, Whole Genome Sequencing (WGS), RNA-Seq processing, GWAS, QTL analysis, gene mapping, structural variation, metagenomics, genome annotation, and phylogenomics (MEGA, Clustal Omega)

Structural Biology, Modeling & Machine Learning: Sequence analysis, primer design, AlphaFold (protein structure prediction), PyMOL (molecular visualization), and foundational familiarity with deep learning frameworks

Infrastructure, HPC & DevOps: Hands-on experience with High-Performance Computing (HPC) clusters, containerization (Docker, Docker Compose, Apptainer/Singularity), and implementation of FAIR Research Data Principles

Research Experience

Karolinska Institute & SciLifeLab, Master's Thesis Research

2026–present

- Investigating replication-stress-induced genomic instability at common fragile sites using targeted long-read sequencing.
- Performing neuroblastoma cell culture, drug treatments, DNA extraction, and long-range PCR.
- Developed bioinformatics workflows using Python, Linux and HPC environments for long-read sequencing analysis.
- Performed structural variant discovery, breakpoint characterization and endogenous barcode identification.

KTH Royal Institute of Technology & SciLifeLab , Research Engineer Sweden - Independently developed Tumor-on-a-Chip models for in vitro anti-cancer drug screening. - Optimized layer-by-layer cellulose nanofibril coatings to enhance cancer spheroid formation. - Contributed to projects on Brain-on-a-Chip models for metabolic studies. - Functionalized electrochemical sensors for ketone detection.	2023–2024
Tehran University of Medical Sciences , Master’s Thesis Research Iran - Synthesized gold nanoparticles conjugated with polyphenols. - Developed microfluidic devices for endothelial cell culture. - Modeled early-stage endothelial dysfunction and conducted oxidative stress assays. - Assisted in preparing a PET/PDMS/Collagen nanofibrous membrane for microfluidic cell culture.	2018–2022
Gen-Iran Lab , Pathology and Cytology Intern Tehran, Iran - Hands-on experience in tissue preparation, fixation, embedding, and sectioning. - Performed histological and immunohistochemical staining.	2023

Wet Lab & Experimental Expertise

Cell Culture & Cancer Biology: 2D & 3D cell culture, tumor spheroid formation and organoid culture, in vitro cancer biology drug screening

Molecular Biology & Genomics Workflows: DNA extraction, downstream cellular and molecular assessments, PCR / Real-Time PCR, agarose gel electrophoresis, Agilent TapeStation analysis, flow cytometry, fluorescent microscopy

Microfluidics & Biosensors: Organ-on-a-Chip modeling, PDMS-based microfluidic systems fabrication, surface and electrochemical sensor functionalization

Histology & Pathology: Histological sample preparation, H&E staining, immunohistochemistry (IHC)

Nanotechnology & Nanomaterial Synthesis: Green synthesis of gold nanoparticles, nanofiber electrospinning, cellulose nanofibrils (CNF) coating optimization

Education

M.Sc. Swedish University of Agricultural Sciences , Bioinformatics - Highest grade (5/5) in Bioinformatics course - Thesis: Exploring the Potential of Fragile Sites as Markers for Lineage Tracing	2025–present
KTH Royal Institute of Technology , Single-Cell & Spatial Transcriptomics Data Analysis Course	2024
M.Sc. Tehran University of Medical Sciences , Medical Nanotechnology Iran - Awarded as the Top Master’s Graduate (GPA: 4/4) - Thesis: Evaluating the Antioxidant Potential of Resveratrol-Gold Nanoparticles in Preventing Oxidative Stress in Endothelium on a Chip	2017–2022
B.Sc. Alzahra University , Cellular and Molecular Biology (Biotechnology Specialization) Iran	2012–2016

Peer-Reviewed Publications

Evaluating the Antioxidant Potential of Resveratrol-Gold Nanoparticles in Preventing Oxidative Stress in Endothelium on a Chip Farzaneh Fayazbakhsh , Fatemeh Hataminia, Houra Mobaleghol Eslam, Mohammad Ajoudanian, Sharmin Kharrazi, Kazem Sharifi, Hossein Ghanbari (Scientific Reports)	2023
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Metabolic Assessment of iPSC-Derived Neurons Under Ketogenic and Normal Diets: Ketone Sensor Development and Comparative Analysis of Neuronal Metabolism * Both authors contributed equally to this work (co-first authors). Rohollah Nasiri*, Farzaneh Fayazbakhsh *, Reyhaneh Sanei, Tingting Wu, Nayere Taebnia, Rouhollah Habibey, Omid Fotouhi, John Cognetti, Volker M. Lauschke, Aman Russom, Anna Herland (CellPress)	2026
Assessing Layer-by-Layer Assembly of Cellulose Nanofibrils in Pancreatic Tumor Spheroid Formation Negar Abbasi Aval, Ekeram Lahchaichi, Oana Tudoran, Farzaneh Fayazbakhsh , Rainer Heuchel, Matthias Löhr, Torbjörn Pettersson, Aman Russom (Biomedicines)	2023
Characterization and Numerical Simulation of a New Microfluidic Device for Cell Investigations H Meslam, F Hataminia, H Asadi-Saghandi, F Fayazbakhsh , N Tabatabaei, Hossein Ghanbari (Frontiers in Lab on a Chip Technologies)	2024
Nanofibers in Respiratory Masks: An Alternative to Prevent Pathogen Transmission Tabatabaei SN, Faridi-Majidi R, Boroumand S, Norouz F, Rahmani M, Rezaie F, Fayazbakhsh F , Faridi-Majidi R (IEEE Transactions on NanoBioscience)	2022

Conference Presentations

Spheroid Formation in Cellulose Nanofibrils-Coated Microfluidic Chips μTAS 2024, Montreal, Canada Poster presentation (first author)	2024
Design of a Microfluidic Electrochemical Ketone Sensor Swedish Microfluidics in Life Science, KTH, Sweden Oral presentation (co-author)	2023
Investigation of Effect of Different Diets on Brain Energy Metabolism Using Organ-on-a-Chip MPS World Summit, Berlin, Germany Poster presentation (co-author)	2023
Preparation of PET/PDMS/Collagen Nanofibrous Membrane Embedded in a Microfluidic Device for Organ-on-Chip Applications The 3rd Conference on Nanofibers, Tehran University of Medical Sciences Poster presentation (co-author)	2022

Professional & Personal Skills

Bridging Wet & Dry Lab: Confident working in both experimental labs and computational environments, making it easy to translate biological questions into data-driven answers.

Independent Problem Solver: Highly autonomous in research environments, with a proven ability to troubleshoot bottlenecks and quickly teach myself new tools to keep projects moving forward.

Collaborative Team Player: Experienced in international, interdisciplinary teams such as KTH, Karolinska Institute and SciLifeLab, with a strong focus on knowledge-sharing and helping colleagues.

Clear Communicator & Fast Learner: Skilled at explaining complex data clearly through peer-reviewed publications and conference presentations; highly adaptable and quick to pick up new scientific topics and skills.

Teaching & Mentorship

KTH Royal Institute of Technology, Intern Supervisor
 Sweden

- Guided an intern in laboratory techniques.

Volunteer Service, Science Teacher

- Taught science to underprivileged children.

Awards & Honors

Top Master's Graduate Tehran University of Medical Sciences	2023
LUSH Prize Candidate Shortlisted for the Young Researcher Award.	2022
Darolfonoon Student Award For voluntary assistance in nanofiber mask production during the COVID-19 pandemic.	2020
Top Article Award First National and Student Conference on Applied Biology, Alzahra University, Iran Prize winner.	2014
Top 5% in Iran's Konkour for Master's Degree University Entrance Iran Ranked 37th in Iran's national Konkour for the Master's Degree University Entrance Exam.	2016

References

Dr. Kasper Karlsson

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